Keyush Shah

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EDUCATION

University of Pennsylvania (School of Engineering and Applied Science) **Philadelphia, PA**
Master of Science in Engineering (MSE) in Data Science; ***GPA 3.9/4.0*** ***Aug 2023 – May 2025***
Coursework: *Machine Learning, Databases, Computer Systems, Deep Learning, Generative Modeling, NLP, Machine Perception*

Kirori Mal College (KMC – University of Delhi) **Delhi, India**
Bachelor of Science (BS) in Statistics; **GPA: 8.11/10** **July 2017 – Aug 2020**

EXPERIENCE

AI Researcher, ***Penn Medicine*** **-** ***Computational Social Listening Lab*** ***Dec 2024 – Present***

Project 1: (**IH Risk Model**)

- Reduced **model tuning time by 10x** by developing an **AutoML** pipeline using sklearn for hernia prediction **(recall: 0.9)**
- Deployed GPT-4 few-shot pipeline on 10k+ redacted surgical notes, **reducing 30% noise vs. BERT** for feature labeling.
- Integrated **vLLM** & **LangChain** for scalable local LLM inference deployed on A100 GPU and **modularized the backend** (OpenAI, Mistral) using **parallelized batch inference** and **memory-aware chunking** across variable length notes.
- Spun up a FastAPI server + TypeScript frontend for RAG workflow, **slashing** surgical data analysis time from **hours to seconds** through ChromaDB vector embeddings and GPT-4 integration.

Project 2: (**Misinformation**)

- Built an NLP pipeline (RoBERTa) to flag health misinformation in ~20k posts, **boosting precision 20%** via entailment.
- Extracted linguistic features from posts using DLATK and applied LDA for topic modeling and performed correlation analysis.
- Consolidated **10K+ survey responses** from multiple platforms into a secure MySQL server and engineered features in **PySpark**.

Data Science Intern (Full-time)**,** ***Universal Media (PA, USA)*** ***May 2024 – Aug 2024***

- Led the development of **3+ data pipelines** using Azure Data Factory, facilitating seamless ingesting into Azure
- Developed python scripts for data transformation, stored them in Blob storage and executed them via batch activity in ADF.
- Drove product insights by building **Mixed Media time series models in Azure Synapse**, analyzing marketing channel impacts on media diversity metrics. Built **Power BI** dashboards to deliver actionable insights for optimizing client strategies.
- **Improved query efficiency by 30%** via stored procedures, parameterization, and indexing of high-frequency access paths.

Assistant Manager (Full-time), ***IIFL Finance Ltd*** ***Apr 2022 – July 2023***

- Analyzed ETL process failures and created **10+** paginated reports in SSIS to help the management track 1000+ branches.
- Optimized & migrated complex SQL queries from an obsolete database server that improved the **reporting services by ~40%.**
- **Digital Adoption:** Led a product-focused initiative to identify digitally savvy customers by engineering features and building ADF pipelines to track campaign behavior. Trained and deployed a Random Forest model (**with a 90% accuracy**) in Azure ML Studio; **exposed it as a REST endpoint consumed by marketing campaigns,** driving digital **disbursal adoption by 50%.**

SELECTED PROJECTS

- **Ride Duration Prediction (2025):** Developed a production-ready ML pipeline to predict NYC taxi ride durations, using **Airflow** for orchestration and **MLflow** for experiment tracking. Achieved **~30% RMSE reduction** via automated hyperparameter tuning & designed modular, reproducible workflows to simulate real-world deployment as a web service via **Flask**. [Link]
- **Multithreaded Image Processing (2025):** Engineered a parallelized box blur algorithm in C++ using POSIX threads, achieving a 2.8× speedup (3251 ms → 1165 ms) on 4 cores by optimizing memory access, leveraging shared-memory synchronization, and **partitioning workloads across non-overlapping thread-local regions**. [Link]
- **FitBit (2024):** Engineered a Django health chatbot leveraging PostgreSQL for robust patient data management, featuring an LLM-agnostic architecture with seamless model switching via **Langchain that reduced overhead by 40%**. Optimized memory usage for long conversations through entity extraction, and **selectively triggered LLM responses** to reduce latency. [Link]
- **Diffusion Transformer (2025):** Implemented PatchVAE with convolutional encoders and patch-based decoding for fine-grained feature extraction. Trained a Diffusion Transformer to sample from the latent space of PatchVAE, achieving a **30% reduction** in FID score and **2x greater feature diversity** compared to VAE-generated samples. [Link]

TECHNICAL SKILLS

Programming Languages: Python, C/C++, SQL, R programming, JavaScript
ML Libraries/Frameworks: PyTorch, scikit-learn, XGBoost, LightGBM, HuggingFace, spaCy, OpenCV, MLflow, A/B Testing
Databases/Web Frameworks: MySQL, PostgreSQL, SSMS, MongoDB, Neo4j, React, NodeJS, Django, Flask, FastAPI, Typescript
Cloud/Big Data Orchestration: AWS (S3, Glue), Azure (Data Factory, Synapse), GCP (BigQuery), Kafka, Airflow, DBT, PySpark
Tools/DevOps: DataBricks, Apache Spark, Docker, Jenkins, Git, Kubernetes, pytest, CI/CD